



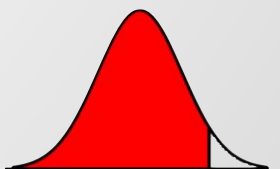
Imbalance-Aware Feature Selection and Machine Learning for U.S. Corporate Bankruptcy Prediction

Yun-Chen Li, Rou-Rou Yu, Yi-An Chen, Huei-Wen Tung



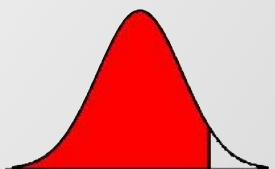
Outline

1. Motivation
2. Study Plan
3. Data (EDA)
4. Model Result
 - LR (Benchmark)
 - XGB
 - SVM
 - RF

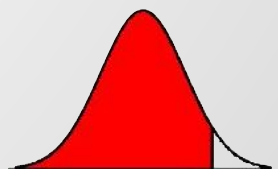
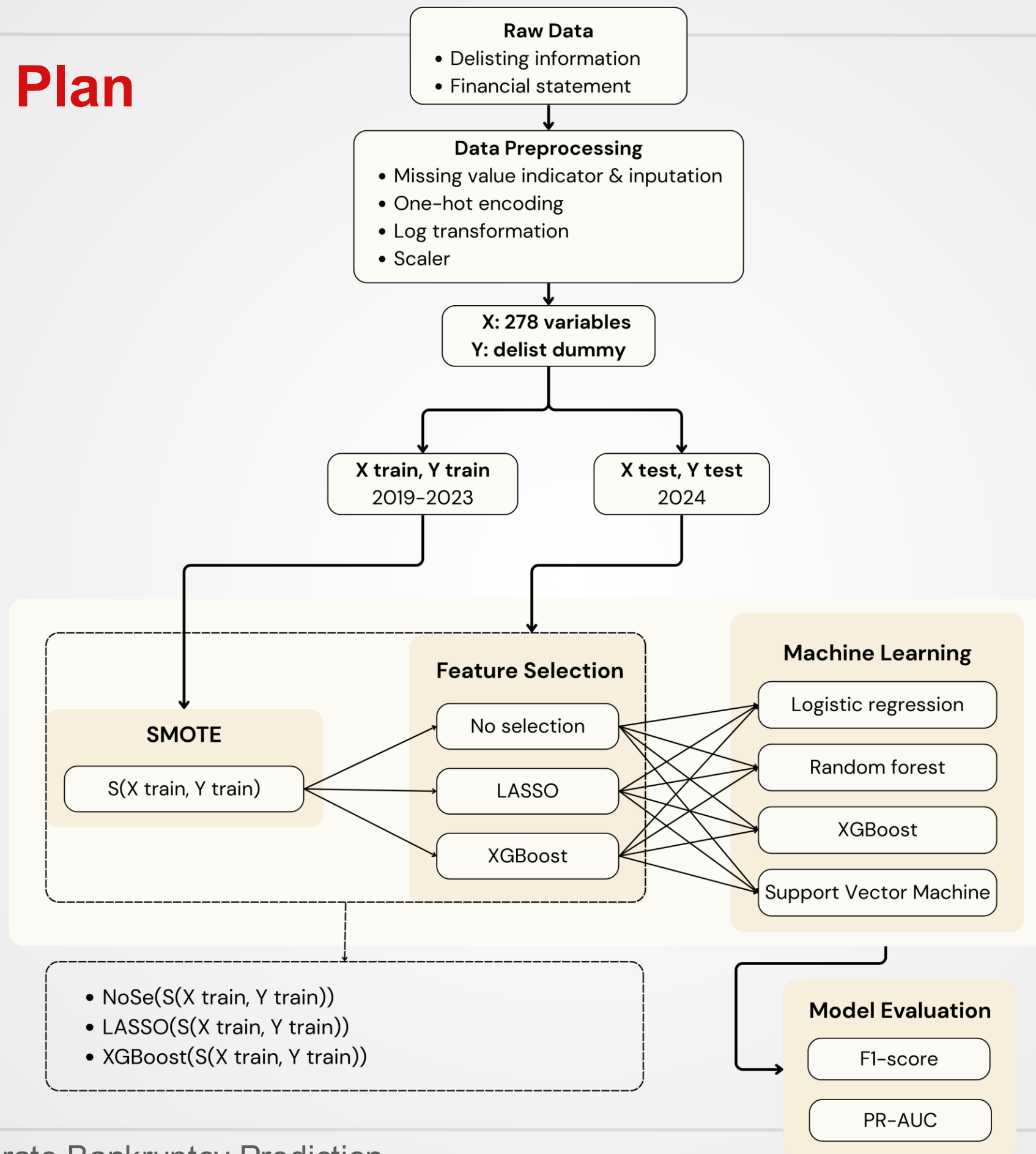


Motivation

- Create a corporate bankruptcy prediction
- Use high dimensional financial data
- Solve class imbalance by creating synthetic samples
- Use feature selection (LASSO, XGBoost)
- Comparing LR, XGB, Random Forest and SVM



Study Plan



Data

- 11475 companies
- 118 features
- Large class imbalance

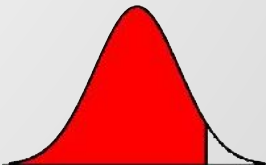
Variable type	Number of type
Categorical	8
Text	6
Continuous	104

Data Information	
Period	2019/1/1-2024/12/31
Frequency	Yearly
Database	
Bankruptcy	WRDS CRSP US Stock delisting information
Financial Statement	WRDS Compustat Fundamental Annual

Delisting Reason Type belongs to BKPY(Bankruptcy)



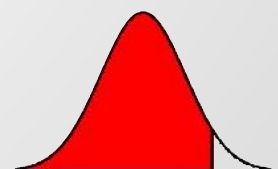
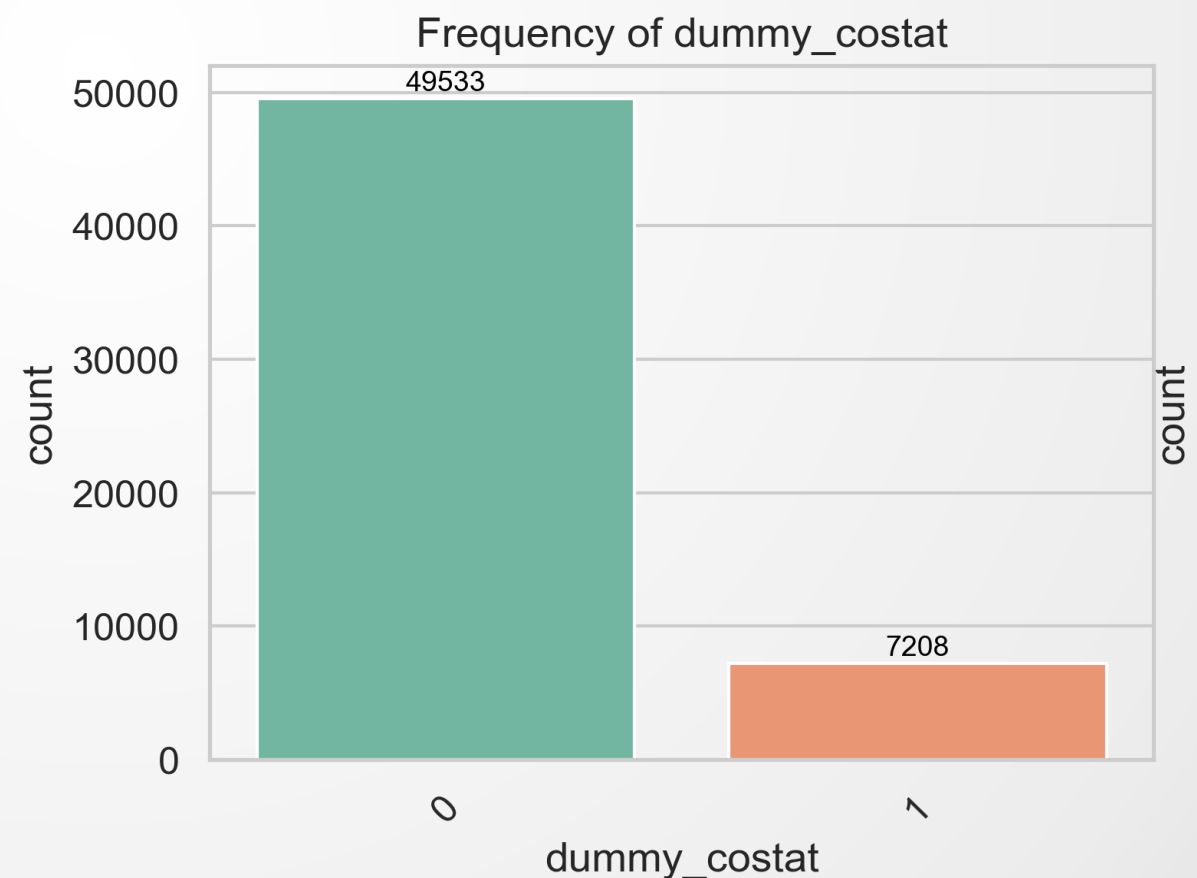
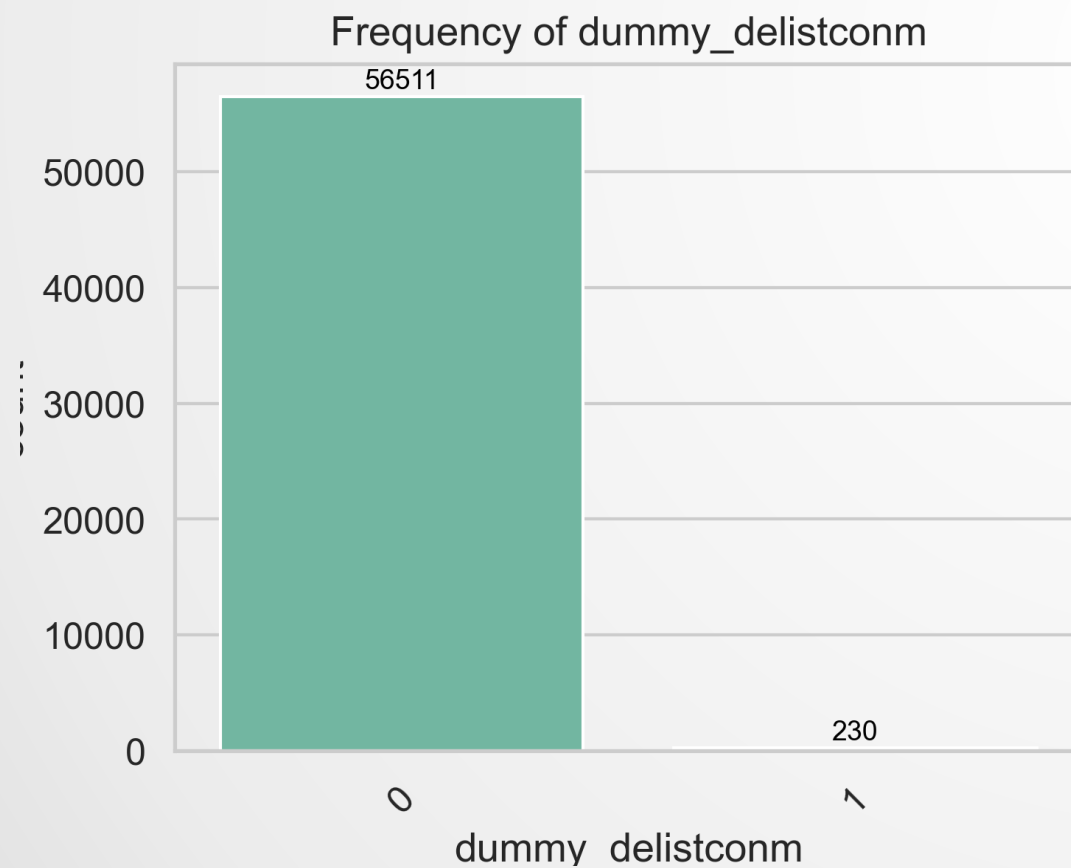
WRDS provides access to standardized, high-quality financial and market datasets that ensure consistency and reliability for empirical research



Data

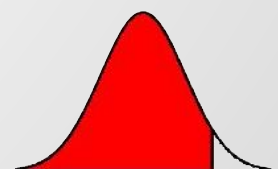
- 11475 companies
- 118 variables
- Large class imbalance

Dummy Variable	Non-Delist Frequency	Non-Delist (%)	Delist Frequency	Delist (%)
dummy delist company	56,511	99.59%	230	0.41%
dummy company status	59,533	87.30%	7,208	12.70%



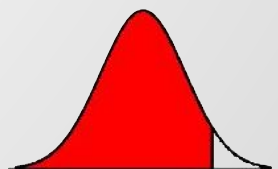
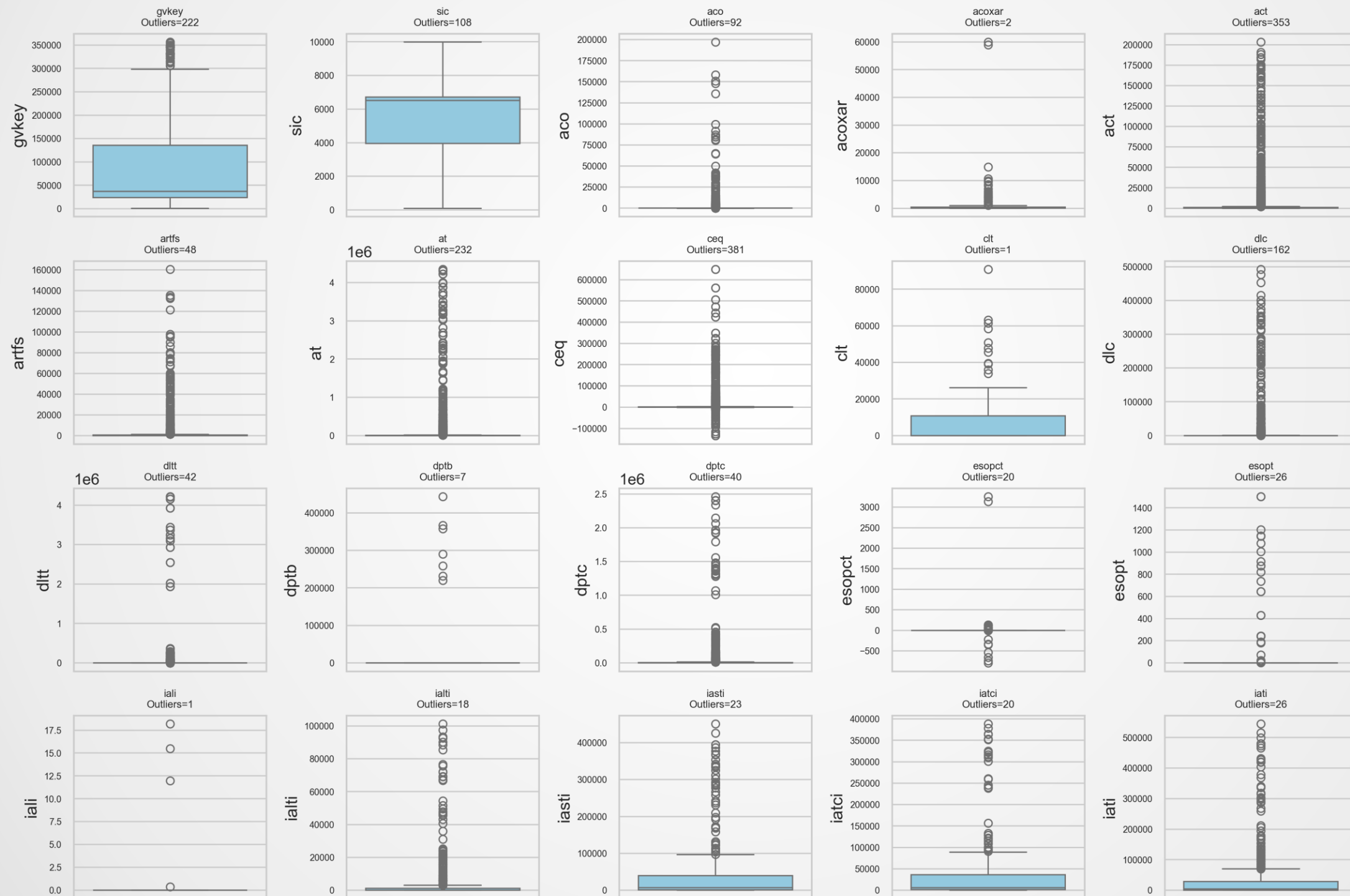
Data pre-processing

Processing types	Effects
Missing value	Fill with use the median for each variable with missing Create a dummy missing indicator for each missing value
One-hot encoding	Create/convert a categorical variable into binary (0/1) feature
Log transforming	To deal with right-skewed distribution
Standard scaler	Uniform Feature Scale



Data

Boxplots of selected features

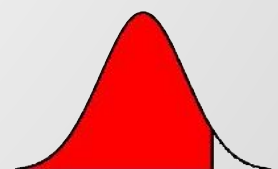


Data

- Description summary for selected numerical features
- Log transformation

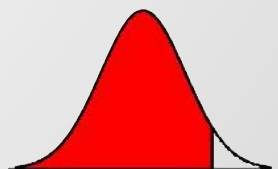
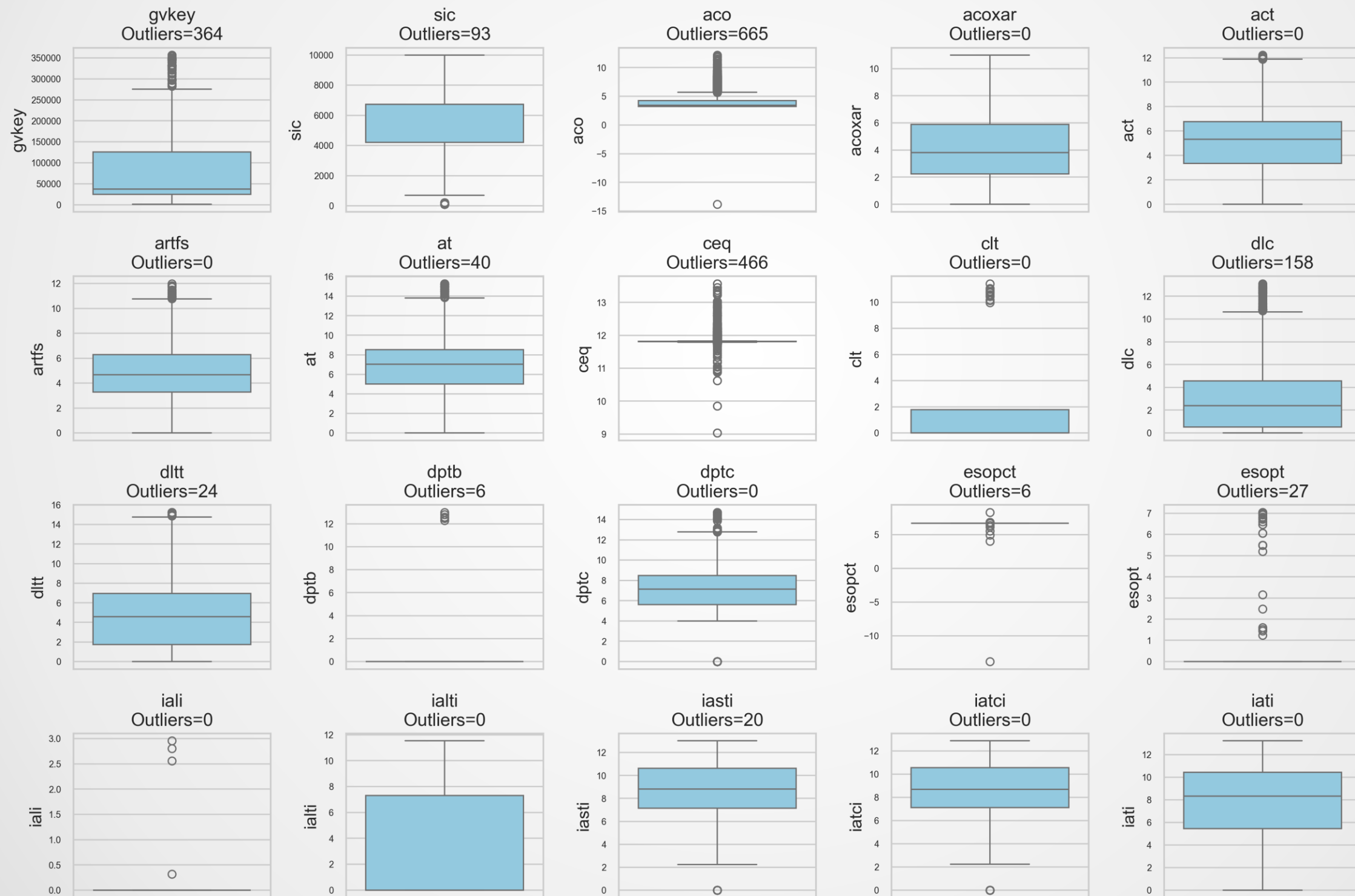
	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
lcat	280.00	7286.79	38020.83	0.00	0.00	0.00	0.00	311448.00	6.29	40.18
dvt	42636.00	156.10	835.28	-709.00	0.00	0.00	36.00	25999.00	12.40	203.35
uopi	1414.00	815.65	1124.10	-3221.00	177.00	395.90	1011.00	9231.00	2.67	9.45
nits	462.00	655.20	1288.71	-3090.00	23.50	144.15	848.05	7713.00	2.36	7.61
artfs	2499.00	2334.35	10414.77	0.00	23.93	104.14	514.04	160673.00	8.11	80.88
ialti	575.00	4833.88	14706.52	0.00	0.00	0.00	1225.19	101294.00	4.38	20.20
lcabg	386.00	5304.75	32527.43	0.00	0.00	0.00	0.00	311448.00	7.44	56.90
cnltbl	527.00	2877.22	8324.58	-0.01	0.00	152.32	1296.13	57646.00	4.32	19.85
ppent	41529.00	1740.90	9394.63	0.00	4.06	37.59	344.87	328806.00	14.04	281.17
pltbl	532.00	10105.54	34816.41	-3354.00	0.00	0.00	1942.50	308810.00	5.16	30.80
idit	25643.00	291.47	4182.15	-16.00	0.00	0.60	15.00	193933.00	26.56	829.10
tstk	42740.00	824.64	6587.93	-26.22	0.00	0.00	7.44	254917.00	17.37	424.41
ptbl	568.00	13957.84	34604.84	-3354.00	453.14	1403.50	8188.25	308810.00	4.71	27.43

The entire description summary refers to overleaf table1



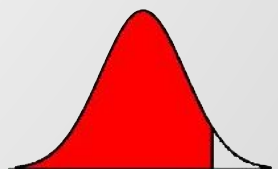
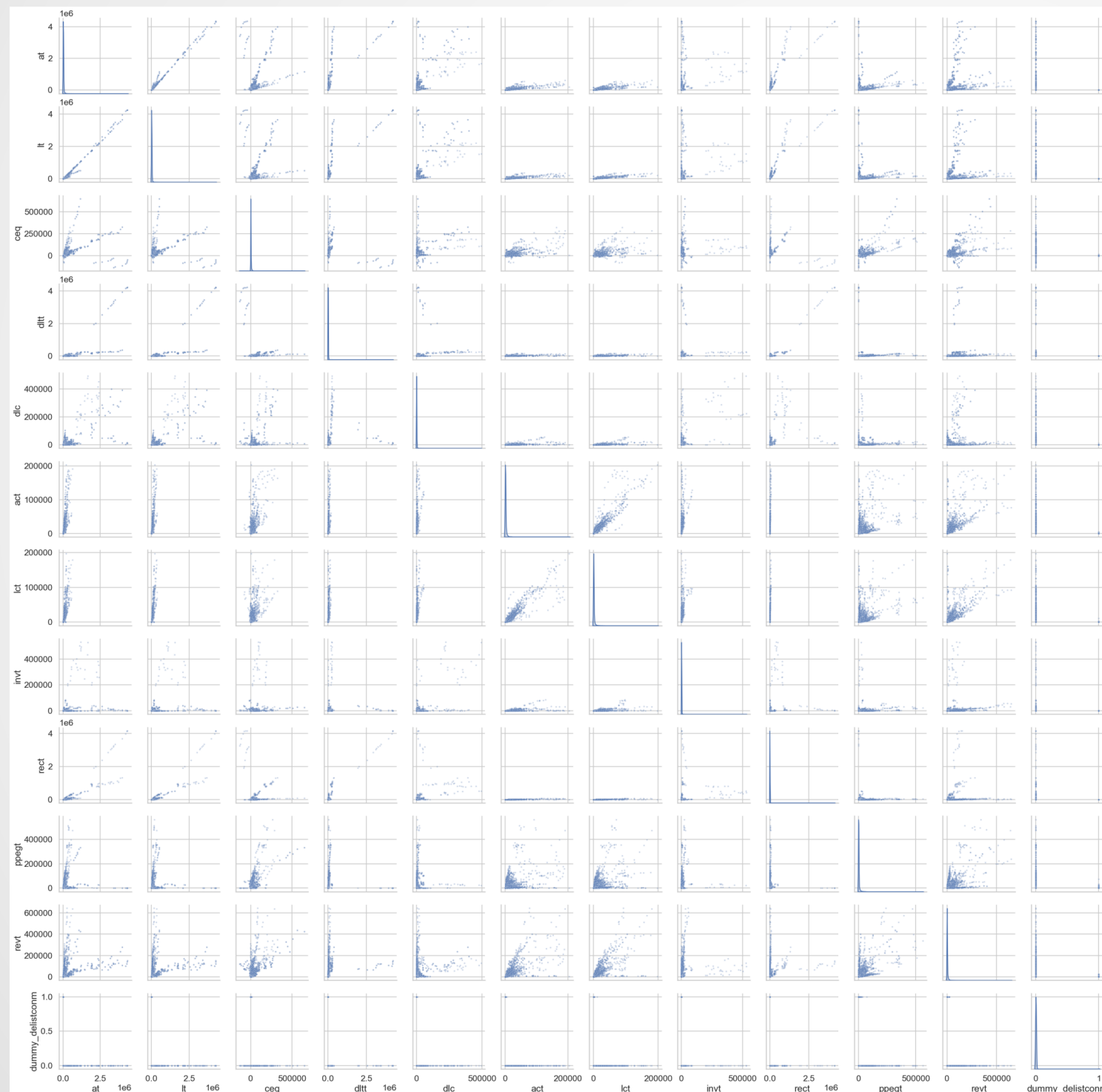
Data

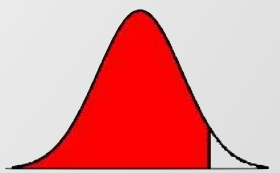
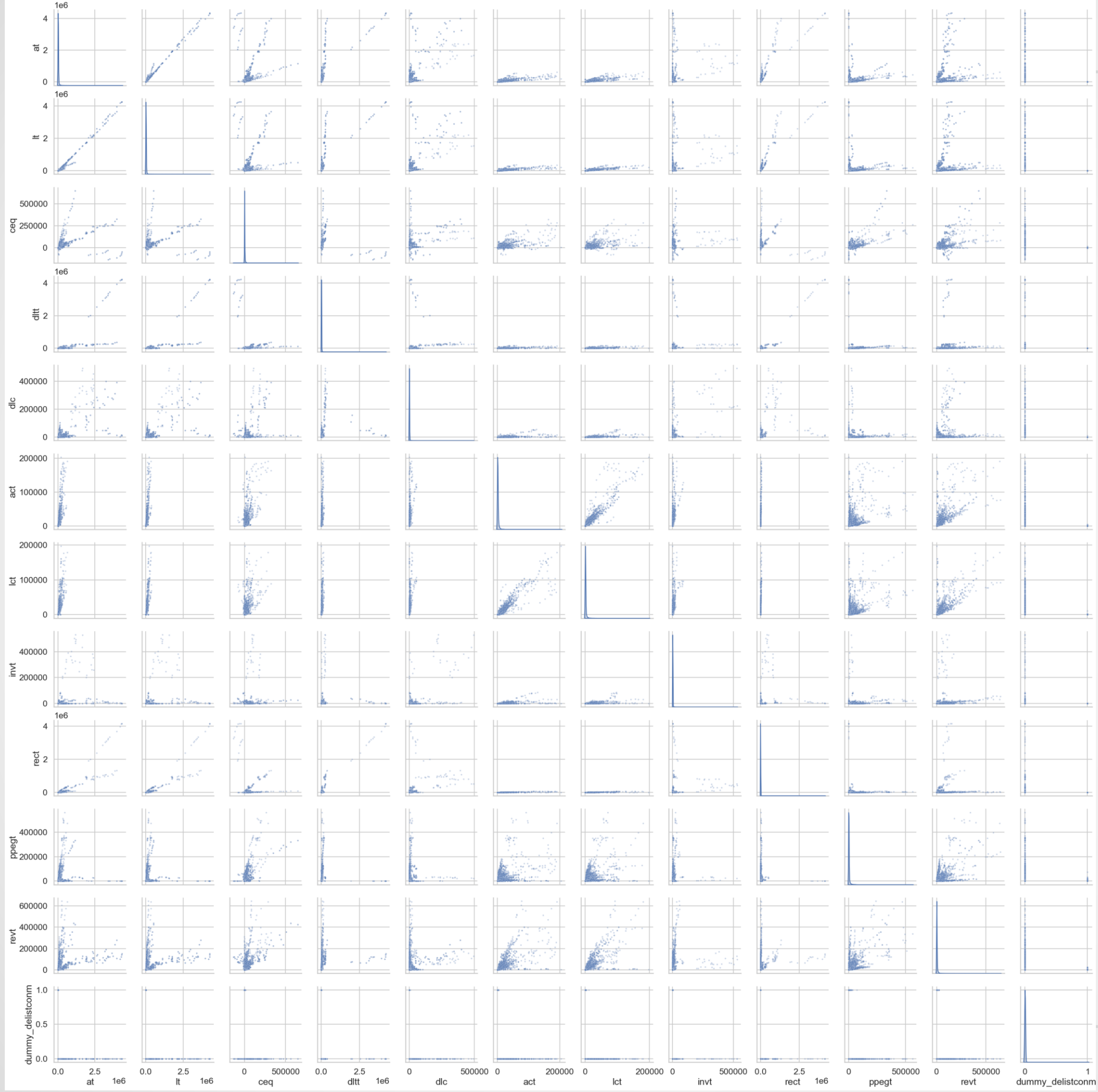
Boxplots of selected features (after log transform)



Data

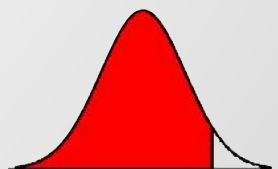
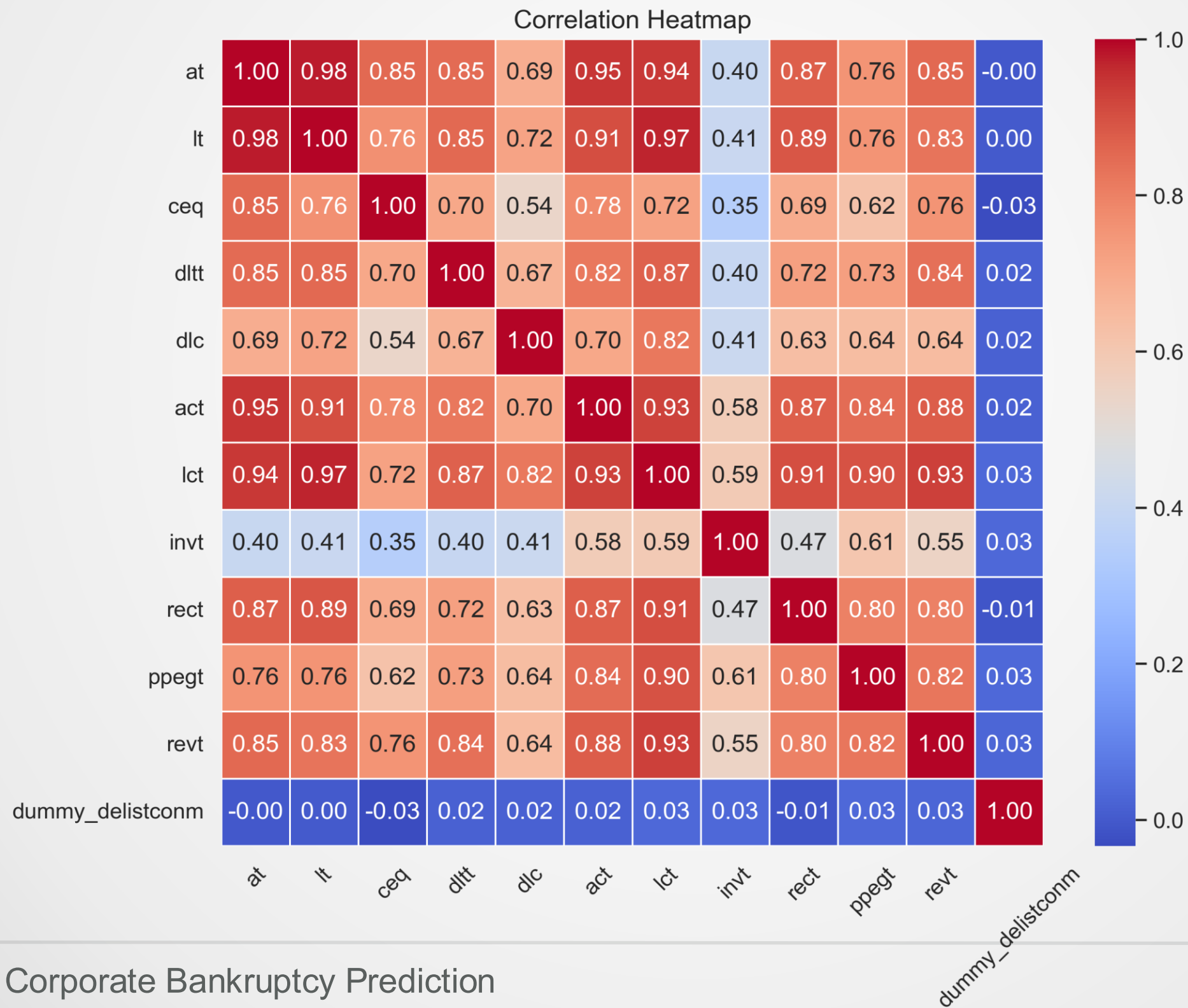
• Pairplot of selected features





Data

Heatmap of selected features



Benchmark

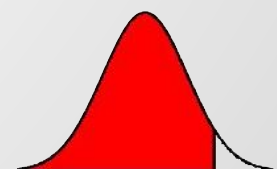
Logistic Regression

- ▣ Couronné, Probst & Boulesteix (2018): Suppose we have observations N , for each observation, let: $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ be the vector of p predictors, and $Y_i \in \{0,1\}$ be the binary outcome. The logistic regression model is defined as:

$$P(Y_i = 1 \mid \mathbf{x}_i) = \pi_i = \frac{1}{1 + \exp \left(-(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \right)}$$

where:

- ▶ π_i is the predicted probability of class 1
- ▶ $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$ is the vector of predictor variables
- ▶ $(\beta_0, \beta_1, \dots, \beta_p)$ are the model coefficients to be estimated



Machine Learning Model

Random forest

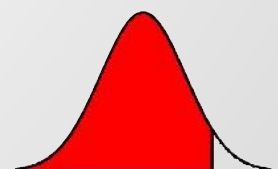
- ▣ Breiman (2001): At a given node t , the Gini impurity is defined as:

$$\text{Gini}(t) = 1 - \sum_{k=1}^K p_k(t)^2$$

where:

- $p_k(t)$ denote the proportion of samples belonging to class k within node t
- ▣ Let $T_b(x_i) \in \{0,1\}$ denote the prediction of the b -th tree for sample i . The prediction is obtained through a majority-voting mechanism across the B trees:

$$\hat{y}_i = \text{mode}\{T_1(x_i), T_2(x_i), \dots, T_B(x_i)\}.$$



Machine Learning Model

XGBoost

- Chen & Guestrin (2016): The model prediction for sample i is obtained by summing the outputs of all regression trees in the ensemble:

$$\hat{y}_i = \sum_{m=1}^M f_m(x_i)$$

where:

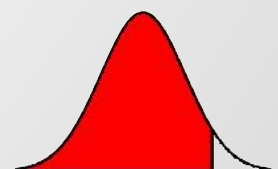
- ▶ $f_m(\cdot)$ denotes the output of the m -th tree
- ▶ M is the total number of trees
- ▶ x_i is the feature vector of the i -th observation

- Apply the sigmoid transformation to convert the score:

$$p_i = \sigma(\hat{y}_i) = \frac{1}{1 + e^{-\hat{y}_i}}$$

where:

- ▶ p_i represents the predicted probability that observation i belongs to the positive class



Machine Learning Model

SVM

- ▣ The model prediction for sample i is obtained by the decision

function: $\hat{y}_i = f(x_i) = \mathbf{w}^T x_i + b$

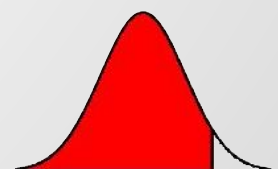
for kernel SVM:

$$\hat{y}_i = \sum_{j=1}^N \alpha_j y_j K(x_j, x_i) + b$$

- ▶ x_i feature vector of the i -th observation
- ▶ $\mathbf{w}$ denotes weight vector
- ▶ b is bias term
- ▶ α_j is lagrange multiplier for support vector j
- ▶ $y_j \in \{-1, +1\}$ is class label
- ▶ $K(x_j, x_i)$ is kernel function
- ▶ N denotes number of support vectors

Apply the sigmoid transformation to convert the score into probability

$$p_i = \sigma(\hat{y}_i) = \frac{1}{1 + e^{-A\hat{y}_i - B}}$$



Feature Selection Model

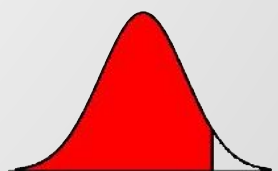
LASSO

- Logistic LASSO: applying an L1 penalty that forces uninformative or redundant coefficients to zero

$$\hat{\beta} = \arg \min_{\beta} \left\{ -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^p |\beta_j| \right\}, \quad p_i = \frac{1}{1 + \exp(-\mathbf{x}_i^T \beta)}.$$

Where:

- ▶ $p_i = \frac{1}{1 + \exp(-x_i^T \beta)}$ denotes the predicted probability
- ▶ $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ is the L_1 norm, and $\lambda > 0$ controls the regularization strength



Feature Selection Model

XGBoost

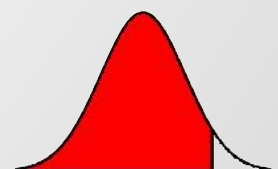
- ▣ XGBoost: the average improvement in the model's objective when the feature is used for splitting.

$$\text{Gain} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right) - \gamma$$

$$\text{Importance}(f) = \frac{1}{S} \sum_{s=1}^S \text{Gain}_s(f).$$

where:

- ▶ λ is the L_2 regularization parameter
- ▶ γ is the penalty for creating a new leaf
- ▶ f is computed as the average gain across all splits where f is used



Results

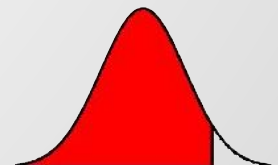
Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9980	0.9769	0.4167	0.2778	0.3333	0.371
SVM	0.9945	0.9948	0.8333	0.1613	0.2703	0.373
RF	0.9853	0.9469	0.1667	0.0147	0.0270	0.113
LR	0.9845	0.9551	0.5833	0.0455	0.0843	0.114

Panel A: LASSO feature selection

Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9982	0.9877	0.6667	0.3636	0.4706	0.476
SVM	0.9954	0.9942	0.7500	0.1765	0.2857	0.335
RF	0.9988	0.9834	0.3333	0.5000	0.4000	0.305
LR	0.9732	0.9430	0.5833	0.0500	0.0773	0.064

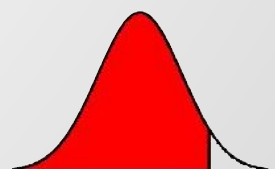
Panel B: XGBoost feature selection

Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9989	0.9832	0.5833	0.5385	0.5600	0.532
SVM	0.9923	0.9836	0.7500	0.1111	0.1935	0.232
RF	0.9990	0.9766	0.2500	0.7500	0.3750	0.289
LR	0.9732	0.9430	0.5833	0.0265	0.0500	0.064



Conclusion

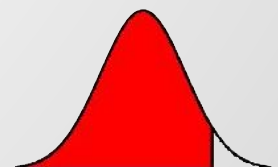
- ▣ XGBoost performs more stable under feature selection and no selection
- ▣ SVM has higher recall but lower precision, and with little improvement after feature selection
- ▣ RF and XGB perform better with feature selection
- ▣ LR has higher AP without feature selections
- ▣ Missing-value indicators disclosure as a distress signal



Model input and output

LASSO feature selection

Symbol	Acronym / Variable name	Description
Y	dummy_delistconm	Equals 1 if the firm experienced a delisting event; 0 otherwise.
X_1	at	Total assets (firm size proxy).
X_2	lt	Total liabilities (financial obligations).
X_3	ceq	Common equity (book value of shareholders equity).
X_4	dltt	Long-term debt (noncurrent portion of total debt).
X_5	dlc	Debt in current liabilities (short-term borrowings).
X_6	act	Current assets (short-term resources).
X_7	lct	Current liabilities (short-term obligations).
X_8	inv	Inventories (goods held for sale or production).
X_9	rect	Accounts receivable (customer credit sales).
X_{10}	ppeg	Property, plant, and equipment (gross).
X_{11}	rev	Total revenue (gross sales).
X_{12}	nit	Net income (profit after taxes).
X_{13}	xopr	Operating expenses (total).
X_{14}	xint	Interest and related expenses (cost of debt).
X_{15}	txt	Income taxes (total).
X_{16}	intan	Intangible assets (goodwill and similar items).
X_{17}	ivst	Short-term investments (marketable securities).
X_{18}	fopt	Funds from operations (cash generated internally).
X_{19}	fsrct	Sources of funds (cash inflows).
X_{20}	fuset	Uses of funds (cash outflows).
X_{21}	dummy_costat	Equals 1 if the firm is classified as inactive in Compustat; 0 otherwise.
$X_{22} X_n$	Industry dummies	Binary indicators for industry sectors (SIC-based).
$X_{n+1} X_p$	Missing indicators	Dummy variables indicating missing accounting items.





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